Assignment 3

Please work on the following questions. Write your answers and paste your charts in a separate word file. (It is preferable that you generate the charts in tableau/power BI)

1. **(30 pts) Sprouts Learning Academy Math Enrichment Program.**
   Sprouts Learning Academy helps prepare students for standardized tests in math for children in the second- through fifth-grade levels. Sprouts assesses each child by giving them a pretest when they start their math enrichment academy program and then again at the conclusion of the academy program. The test results are measured in percentiles compared to all other students taking similar tests. (Use the file ***SproutsLearningChart*** ***from Chapter 7 in dataset.zip)***

   a. Create **a clustered column chart** that compares the results of these pre- and post-academy tests for each grade level.
   b. Create **a slope chart** that compares the results of these pre- and post-academy tests for each grade level. Which chart (cluster column chart or slope chart) is better for telling the story about how well the math enrichment program is doing in improving performance on this standardized test?
   c. Test scores have increased for all grade levels between pre- and post-math-enrichment program tests. However, test scores for fourth-grade level have less difference than the test scores of second-, third-, and fifth-grade levels. Use the **preattentive attribute of color** for **above slope chart** to emphasize the difference in the results for fourth-grade level compared to second-, third-, and fifth-grade levels.

2. **(70 pts) Building a Donor Data Dashboard**. The American Retriever Foundation (ARF) is a not-for-profit organization dedicated to health issues faced by the six distinct retriever breeds (Chesapeake Bay, curly-coated, flat-coated, golden, Labrador, and Nova Scotia duck tolling). ARF needs to develop a data dashboard to monitor its donor activity and its interactions with potential donors. Management is concerned primarily with the number and dollar value of donations, the number of legacy donors (those who have donated in the past twelve months) and new potential donors solicited, and the number of solicitations that result in donations. They want to compare these results across ARF's four development officers (Randall Shalley, Donna Sanchez, Marie Lydon, and Hoa Nguyen) by date and mode of contact (telephone, email, or personal meeting).
   (Use the **data tab** in the file ***ARFCharts*** ***from Chapter 8 in dataset.zip)***

   ARF has collected data for each solicitation initiated last year from its relational database. These data include the solicitation number, development officer, date of solicitation, mode of solicitation, whether the solicitation resulted in a donation, and whether the solicited potential donor was a legacy donor. ARF also added a field for month in which the solicitation was made.

a. Please create the following **chart and slicing filters worksheets** for the data dashboard in question c: (**check this link to create slicer in tableau**)

1) A stacked bar chart of the number of solicitations across development officer by whether the solicitation resulted in a donation. Add a slicing filter for development officer, a slicing filter for whether the solicitation resulted in a donation, and a slicing filter for month of solicitation in this chart.
2) A stacked bar chart of the percentage of successful solicitations across mode of solicitation by whether the solicitation resulted in a donation. Add a slicing filter for mode of solicitation, a slicing filter for whether the solicitation resulted in a donation, and a slicing filter for development officer in this chart.
3) A stacked bar chart of total donations (in $1000s) across development officer by legacy designation. Add slicing filters for development officer, legacy status of the donor, and month of solicitation in this chart.
4) A cluster bar chart of total donations (in $1000s) across mode of solicitation by legacy designation. Add slicing filters for mode of solicitation, legacy status, and month of donation in this chart.

b. Create a data dashboard and naming it *Dashboard*; moving above worksheets that containing charts and slicing filters to the *Dashboard*; repositioning the charts on the *Dashboard*; and adding a title to the dashboard and doing whatever formatting and editing is necessary to make the dashboard functional and visually appealing.

c. Amend the **data dashboard** you created in part b in the following ways. (**check this link for applying one slicer filter to multiple worksheets**)

1) Create a single slicer to filter month for the charts created in the *a1, a2 and a4*.
2) Create a single slicer to filter development officer for the charts created in the *a1, a2 and a3.*
3) Create a single slicer to filter whether or not a solicitation resulted in a donation for the charts created in a1 and a2.
4) Create a single slicer to filter legacy status for the charts created in a3 and a4.
5) Create a single slicer to filter mode of solicitation for the charts created in the *a2 and a4.*

Once you have amended the data dashboard, rearrange the charts and slicers to create an effective and visually appealing dashboard. Test each slicer to ensure it works on the appropriate charts.

d. The file ***ARFNewData*** contains 50 entries for the past year that were not originally entered into the relational data base before the data for this dashboard was retrieved. Add these data to the ***ARFData*** table and refresh all **charts**. Comment on differences between **the resulting dashboard** and **the dashboard from part b**.

General Submission Instructions:
All work must be your own. Copying other people's work or from the Internet is a form of plagiarism and will be prosecuted as such.

You may submit a Microsoft Word (.doc/.docx) document as an attachment using the Canvas Assignment tool, or you may copy and paste your answer into the provided box within the Assignment tool. If you attach a document for your assignment, be sure to include your name in the text of the document and in the name of the document.

- You can only submit once, so make sure you are completely finished before submitting and that you attach the correct word .doc/.docx file.
- Submissions sent by email will NOT be accepted.

Due dates are listed in the Assignment Schedule document.